

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Heat Recovery Steam Generator IE-HR3 - Datasheet

Document number	IEC-ALP-DS-104
Revision	B
Date	2026-09-28
Status	Revised for EPC engineering - supersedes Rev 0
Addressee	Istanbul EPC Muhendislik ve Taahhut A.S. (consortium leader) - EPC engineering
Contract	ALP-EPC-001 (Owner contract); consortium agreement IEPC-IEC

	Prepared	Checked	Approved
Function	IEC plant integration engineer	IEC lead engineer (discipline)	IEC project engineering manager
Signature / date	signed 2026-09-28	signed 2026-09-28	signed 2026-09-28

Rev	Date	Description
0	2026-09-27	Issued for EPC engineering (reference data)
A	2026-09-28	HRSG arrangement and thermal data corrected (TQ-IEC-016)
B	2026-09-28	SCR and ammonia injection sized for LDO with GT-outlet NOx 150 mg/Nm3 (TQ-IEC-018)

CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. General

Item	Data
Type	IE-HR3 horizontal gas flow, triple-pressure reheat, natural circulation (HP drum thin-wall design with external cyclone separators), modular (22 modules), self-supporting casing, internally insulated
Design code	ASME Section I, PED 2014/68/EU module G (notified body)
Cycling	250 starts/year, 30-year fatigue life per EN 12952-3 (hot 180, warm 60, cold 10 per year)
Gas side pressure loss at SRC	30 mbar (incl. SCR and CO catalyst, clean)
Outlet to stack	HRSG outlet transition duct flange, 9.2 m diameter (stack by EPC, IF-26)
SCR	V2O5/TiO2 honeycomb at 360 - 370 degC (between IP superheater and HP economiser 2), aqueous ammonia 24.5 % injection grid (IEC); design NOx at HRSG outlet 22 mg/Nm3 (15 % O2); NH3 slip <= 3 mg/Nm3; catalyst life 24,000 h
CO catalyst	Pt/Pd oxidation catalyst at 450 degC (after HP superheater/reheater)
Aqueous ammonia consumption	125 kg/h at base load on gas; 480 kg/h on LDO (GT-outlet NOx 150 mg/Nm3, 67 % removal, TQ-IEC-018)
Stack damper	not included (stack by EPC)

2. Thermal data at SRC-NG-100

Rev B (TQ-IEC-018): SCR catalyst volume increased by 35 % and ammonia injection grid resized for LDO operation with GT-outlet NOx up to 150 mg/Nm3 (water/fuel ratio 0.60); ammonium bisulphate protection: minimum SCR inlet temperature 330 degC on LDO.

Circuit	Flow kg/s	Outlet pressure bar(a)	Outlet temperature degC	Drum pressure bar(a)	Design pressure bar(a)
HP superheater	100.15	187.0	600.0	195.0	210
Reheater	117.49	37.6	600.0	-	50
IP superheater	17.34	40.3	330.0	41.8	50
LP superheater	11.90	5.40	250.0	5.9	10
Condensate preheater	141.87	6.9	153.2	-	25

Section	Gas in degC	Gas out degC	Duty MW
HPSH / RH	640.0	450.6	176.83
HPEVAP	450.6	371.6	71.38
IPSH	371.6	366.9	4.19
HPECO2	366.9	295.1	63.71
IPEVAP	295.1	261.0	29.89
LPSH	261.0	258.2	2.43
HPECO1 / IPECO	258.2	198.7	51.64
LPEVAP	198.7	166.2	27.91
CPH	166.2	80.2	73.12

Stack temperature 80.2 degC at SRC. Condensate preheater inlet temperature is kept at not less than 50 degC (acid dew point margin, LDO) by a 2 x 100 % CPH recirculation pump (IEC); at SRC the mixed inlet from the condensate and the fuel gas heater return is 30.9 degC before recirculation.

3. Water/steam quality required

Per IEC-ALP-REQ-001 section 4 (IAPWS TGD and VGB-S-010): all-volatile treatment (AVT(O)) with trisodium phosphate provision for the HP drum; continuous blowdown 0.5 % maximum at start-up, 0 % in normal operation.

4. Weights and transport

Item	Data
Heaviest pressure-part module (HP evaporator)	185 t, 24.0 x 3.6 x 26.5 m (shipped horizontal)
HP drum	96 t, 14.0 m long, OD 1.9 m
Total HRSG weight (pressure parts, casing, steel, catalysts)	6,400 t